

Project ID:

R26-CS-016

1. Topic (12 words max)

Multi-Layer Dos Detection and Severity Scoring System Using Hybrid ML and DL Approaches.

2. Research group the project belongs to

IAS - Information Assurance & Security

3. Specialization of the project belongs to

Cyber Security (CS)

4. If a continuation of a previous project:

Project ID	R26-CS-016
Year	2026

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Denial of Service (DoS) attacks remain a critical cybersecurity threat that disrupts network availability by overwhelming systems with malicious traffic. Organizations like enterprises, cloud service providers, ISPs, financial institutions, healthcare systems, and government networks are primary victims due to their dependence on continuous network availability. Current DoS detection systems largely operate on binary outcomes (attack vs. normal) and fail to differentiate the relative severity of attack types, leading to inefficient prioritization of mitigation actions. Network administrators and SOC analysts (future users of the solution) require an intelligent system that not only detects multi-layer DoS attacks but also quantifies their real-world impact to support fast and informed decision-making. [1], [2]

Technically, unresolved DoS attacks cause service outages, bad performance, and resource exhaustion across Layer 3 and Layer 4 protocols [2]. Economically, long downtime results in financial losses, SLA violations, and reputational damage [4]. Socially, disruptions in critical services such as healthcare and banking affect public trust. From a legal and ethical perspective, failure to detect and respond to high-severity attacks may lead to regulatory non-compliance and negligence claims, especially in critical infrastructure sectors. [5]

This research fits with the Information & Security (IAS) research cluster, focusing on securing digital infrastructures through intelligent and adaptive security mechanisms. The proposed system integrates Machine Learning and Deep Learning techniques to enhance network threat detection, analysis, and risk assessment in multi-layer environments. Our research addresses contemporary challenges in information assurance by improving the accuracy and contextual understanding of DoS attack severity. Relevant industry verticals include cloud computing platforms, telecommunications networks, financial services, healthcare IT systems, and smart infrastructure environments, where continuous availability and robust security monitoring are critical. [2], [5]

The proposed system supports SDG 9 (Industry, Innovation, and Infrastructure) by strengthening resilient digital infrastructure and SDG 16 (Peace, Justice, and Strong Institutions) by enhancing cyber resilience against disruptive attacks. By enabling efficient threat prioritization, it promotes sustainable and secure digital ecosystems. [5]

[1] I. R. Sabu *et al.* [1] I. R. Sabu *et al.*, "Detection of DoS Attacks Using Machine Learning Based Intrusion Detection System," *IEEE InC4*, 2024.

[2] D. V. Hernandez *et al.*, "Real-Time DDoS Detection in High-Speed Networks," *Electronics*, 2025.

[3] N. H. Syafiuddin *et al.*, "Detection of SYN Flood and UDP Lag Attacks Using AdaBoost," *IEEE ICoDSA*, 2023.

[4] F. W. Romadhon *et al.*, "Comparative Analysis of HAProxy and Nginx in Mitigating UDP Flood Attacks," *IEEE ICoICT*, 2024.

[5] S. Sutradhar and J. L. Sarkar, "A2M: Adaptive Multi-Layered Defence Mechanism Against DDoS Attacks," *Int. J. Applied Mathematics*, 2025. [2] D. V. Hernandez *et al.*, [1] I. R. Sabu *et al.*,

6. Brief description of Existing Research and Systems (300 words max)

Most of the Existing DoS mitigation systems largely rely on traditional Intrusion Detection Systems (IDS), load balancers, and rule-based or statistical defenses. Commercial and open-source tools such as Suricata IDS, HAProxy, and Nginx are commonly deployed to mitigate volumetric attacks like UDP flooding by distributing traffic and applying threshold-based filtering. Comparative studies says that HAProxy performs better than Nginx in handling high-volume UDP flood attacks due to its event-driven architecture and lower resource consumption, while IDS tools are effective mainly for detection rather than prioritization or severity assessment

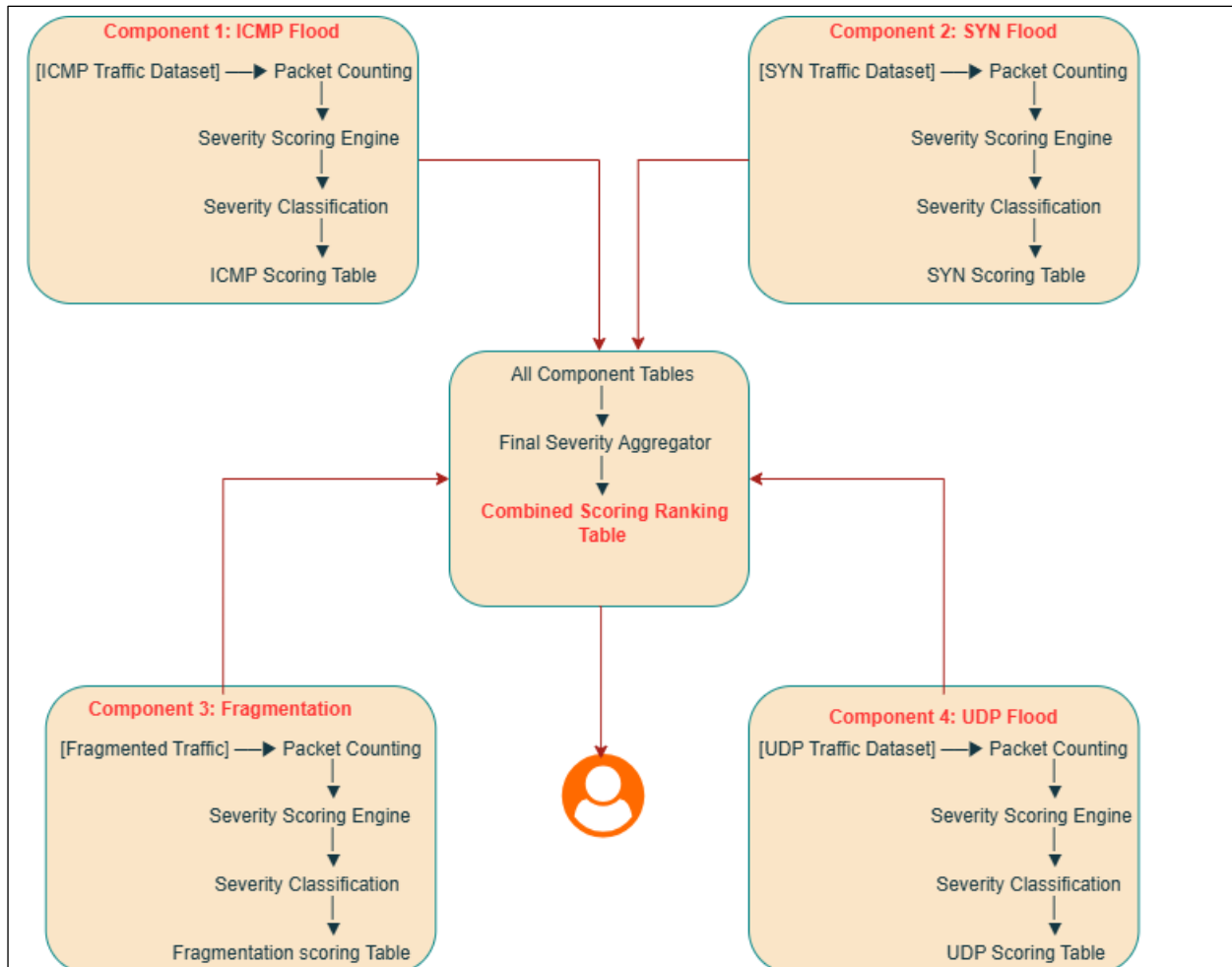
Comparative_Analysis_of_HAProxy.

However, these systems focus primarily on mitigation or binary detection, lacking intelligent severity differentiation.

Recent research extensively how a machine learning (ML) and deep learning (DL) for DoS detection. ML-based IDS solutions using **SVM**, **Random Forest**, **AdaBoost**, and ensemble techniques demonstrate high detection accuracy for attacks such as SYN flood and UDP flood. Deep learning approaches using **LSTM**, **CNN**, **TCN**, and **hybrid CNN-LSTM** models more enhance detection performance in real-time and high-speed networks. Multi-layered defense architectures have been proposed to detect attacks across network, transport, and application layers, improving resilience against multi-vector DoS attacks.

Even with high detection accuracy, existing research primarily treats DoS detection as a binary or multi-class classification problem without quantifying attack severity or impact level. Most systems do not differentiate the relative danger of attack types (e.g., SYN flood vs ICMP flood), limiting effective response prioritization. Additionally, many architectures focus either on detection or mitigation, lacking an integrated severity-aware decision mechanism. This gap highlights the need for a weighted severity scoring system that complements multi-layer detection to support faster, risk-informed mitigation decisions.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).



The proposed solution addresses the key limitations of existing DoS detection systems by extending over binary or multi-class attack detection. While earlier systems accurately identify attack types, they lack a mechanism to quantify the relative severity and real-world impact of different attacks. This research introduces a severity-aware scoring system integrated with multi-layer DoS detection. By assigning impact-based weights to attack types such as SYN flood, UDP flood, fragmentation attacks, and ICMP flood, the system enables risk-prioritized decision-making. This approach allows security operators to identify not only the existence of attacks but also their urgency, enabling faster and more effective mitigation.

System Perspective: Input – Process – Output

Input

Real-time or offline network traffic

Layer 3 and Layer 4 packet features (ICMP, TCP SYN, UDP, fragmentation)

Packet rates, protocol flags, and time-window statistics

Process

1. Traffic Capture & Feature Extraction

Network traffic is monitored and relevant L3/L4 features are extracted using time-window analysis.

2. Multi-Layer Attack Detection

A hybrid Machine Learning + Deep Learning model identifies attack presence and classifies attack types.

3. Attack Quantification

Detected attack packets are counted separately for ICMP, SYN, UDP, and fragmentation attacks.

4. Severity Scoring Engine (Core Contribution)

Each attack count is multiplied by a predefined impact-based weight:

- SYN Flood (3.0)
- UDP Flood (2.5)
- Fragmentation Attack (2.0)
- ICMP Flood (1.5)

5. Severity Classification

The total score is mapped to severity levels (Low, Medium, High, Critical).

Output

- Detected attack type(s)
- Individual attack severity scores
- Total severity score
- Overall severity level to guide mitigation priority

8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

The project team can gain essential domain expertise through a combination of academic literature review, hands-on experimentation, and guided supervision. Existing IEEE and journal publications on DoS detection, multi-layer security architectures, and machine learning-based intrusion detection systems provide a strong theoretical foundation. Additionally, consultation with academic supervisors and industry practitioners in cybersecurity and network security will support deeper understanding of attack behaviors, protocol-level vulnerabilities, and real-world mitigation practices. Online platforms, workshops, and cybersecurity labs further contribute to practical exposure.

The technical skills required for this research project include network security fundamentals, machine learning and deep learning techniques, traffic analysis, and data processing. Team members can acquire these skills through structured coursework, online learning platforms, and hands-on implementation using tools such as Python, and network analysis utilities (e.g., Wireshark). Practical experimentation with publicly available datasets and simulation environments enables members to enhance their understanding of feature extraction, model training, evaluation, and performance analysis. Continuous experimentation and iterative model refinement further enhance technical competence.

This project basically relies on publicly available benchmark datasets widely used in cybersecurity research. Datasets such as CICDDoS2019, CAIDA, and similar repositories provide labeled traffic data covering multiple DoS attack types, including SYN flood, UDP flood, ICMP flood, and fragmentation attacks. These datasets are suitable for training, validation, and testing of the proposed hybrid ML/DL models and severity scoring mechanism. If required, additional traffic can be generated in controlled environments using traffic simulation tools to validate model robustness, ensuring no real-world systems are impacted.

Our project does not require formal ethical clearance as it does not involve human participants, personal data, or sensitive identifiable information. All experiments are conducted using synthetic or publicly available anonymized datasets and controlled simulations. No real-world attack execution against live systems is performed. This research strictly adheres to ethical guidelines by focusing on defensive cybersecurity mechanisms, responsible data usage, and compliance with institutional research policies.

9. Objectives and Novelty

Main Objective

This system gives a unified multi-layer Denial-of-Service (DoS) detection and severity scoring framework that integrates hybrid Machine Learning and Deep Learning techniques with an impact-based weighted scoring mechanism to accurately detect, classify, and prioritize Layer 3 and Layer 4 DoS attacks, including ICMP flood, SYN flood, UDP flood, and fragmentation attacks. The framework operates on real-time network traffic collected at packet and flow levels and extracts statistical, protocol-specific, and temporal features such as traffic rates, flag distributions, fragmentation behavior, entropy metrics, and burst patterns.

Apart from detection, the system performs attack-type classification and computes a dynamic severity score using an impact-based weighted model that accounts for traffic intensity, attack persistence, protocol criticality, resource exhaustion potential, and service impact. This severity score enables risk-aware prioritization and supports real-time mitigation decisions by aligning technical attack characteristics with operational risk, thereby improving response accuracy, scalability, and resilience against both high-rate and stealthy DoS attacks.

Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22090126 KOWSHICK S N	ICMP Flood Detection & Severity Analysis	• Extract ICMP traffic features (packet rate, echo requests) • Perform ICMP packet counting • Apply weighted severity scoring (Weight = 1.5) • Classify ICMP attack severity levels • Generate ICMP attack ranking table	Introduces severity-aware ICMP flood analysis , moving beyond binary detection by quantifying ICMP attack impact using weighted scoring

IT22284334 FARHAN R	SYN Flood Detection & High-Impact Scoring	• Detect TCP SYN flood traffic • Count SYN packets and connection attempts • Apply high-impact severity weighting (Weight = 3.0) • Classify severity and rank SYN attacks	Prioritizes connection-exhaustion attacks using impact-based scoring, which is rarely addressed explicitly in existing DoS research
IT22284334 KEZOTHRAN G	Fragmentation Attack Detection & Analysis	• Identify fragmented IP packets • Perform fragmentation packet counting • Apply medium-impact severity weighting (Weight = 2.0) • Classify fragmentation attack severity	Provides explicit fragmentation attack severity quantification , which is often ignored or merged with generic DoS detection in prior work
IT22122346 KAJAN S	UDP Flood Detection & Aggregation	• Detect high-volume UDP traffic • Perform UDP packet burst analysis • Apply severity weighting (Weight = 2.5) • Generate UDP attack severity table • Integrate all component outputs into final severity ranking	Combines multi-attack severity tables into a unified ranking model , enabling comparative attack prioritization rather than isolated detection

10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

- a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes		No	
-----	--	----	--

- b) The proposed topic exhibits novelty.

Yes		No	
-----	--	----	--

- c) The student group can successfully execute the proposed project.

Yes		No	
-----	--	----	--

- d) The proposed sub-objectives reflect the students' areas of specialization.

Yes		No	
-----	--	----	--

- e) Any other comments:

--

	Title	First Name	Last Name	Signature
Supervisor				
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				